Online Hospital Appointment Booking System

Aim

To design and develop an online hospital appointment booking system that lets patients search hospitals and doctors, book and cancel appointments and write reviews, and lets hospitals and the admin manage doctors, appointments and statistics.

Algorithm

1. Start the application; the user or hospital registers and logs in.
2. Hospital creates its details, then updates details and location, and adds doctors with their schedules.
3. User searches hospitals by location or doctors by specialty and views the hospital list and Google map.
4. User selects a doctor, views the available slots and books the appointment.
5. User fills the check-in form and receives an email or SMS reminder.
6. User views the appointment status and past appointments, cancels if required and writes a review for the doctor.
7. Hospital manages appointments and doctor timings.
8. Admin manages doctors and appointments, views statistics and ratings, and logs out.

CODE:

HTML

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>MedCare Hospital</title>

    <link rel="stylesheet" href="Hospital.css">

</head>

<body>



<header>
```

```
    <div class="logo">✚ MedCare</div>
    <nav>
      <a href="#home">Home</a>
      <a href="#doctors">Doctors</a>
      <a href="#booking">Book Appointment</a>
      <a href="#history">My Appointments</a>
      <a href="#feedback">Reviews</a>
      <a href="#admin">Admin</a>
    </nav>
</header>

<section class="hero" id="home">
    <div class="hero-content">
      <p class="tag">YOUR HEALTH, OUR PRIORITY</p>
      <h1>Better Care.<br>Better Health.</h1>
      <p>Find trusted doctors and book your hospital appointment easily.</p>
      <a href="#booking" class="hero-btn">Book an Appointment</a>
    </div>
</section>

<section class="container" id="search">
    <h2>Find a Doctor</h2>
    <p class="subtitle">Search by specialty or hospital location.</p>

    <form id="searchForm" class="search-grid">
```

```
<div>
    <label>Specialty</label>
    <select id="searchSpecialty">
        <option value="">All Specialties</option>
        <option>General Physician</option>
        <option>Cardiologist</option>
        <option>Dermatologist</option>
        <option>Pediatrician</option>
        <option>Dentist</option>
        <option>Orthopedic</option>
    </select>
</div>

<div>
    <label>Location</label>
    <select id="searchLocation">
        <option value="">All Locations</option>
        <option>Chennai</option>
        <option>Coimbatore</option>
        <option>Bengaluru</option>
        <option>Hyderabad</option>
    </select>
</div>

<div>
```

```
            <label>Doctor Name</label>
            <input type="text" id="searchName" placeholder="Enter doctor name">
        </div>

        <button type="submit" class="primary-btn">Search Doctors</button>
    </form>

    <div id="doctorResults" class="doctor-grid"></div>
</section>

<section class="container" id="doctors">
    <h2>Our Doctors</h2>
    <p class="subtitle">Choose the right specialist for your healthcare needs.</p>
    <div id="allDoctors" class="doctor-grid"></div>
</section>

<section class="container" id="account">
    <h2>Patient Account</h2>
    <p class="subtitle">Register or log in to manage your appointments.</p>

    <div class="two-columns">
        <form id="registerForm" class="form-card">
            <h3>Patient Registration</h3>

            <label>Full Name</label>
```

```
    <input type="text" id="regName" required>

    <label>Email</label>
    <input type="email" id="regEmail" required>

    <label>Password</label>
    <input type="password" id="regPassword" minlength="4" required>

    <button class="primary-btn" type="submit">Register</button>
</form>

<form id="loginForm" class="form-card">
    <h3>Patient Login</h3>

    <label>Email</label>
    <input type="email" id="loginEmail" required>

    <label>Password</label>
    <input type="password" id="loginPassword" required>

    <button class="primary-btn" type="submit">Login</button>
    <button class="secondary-btn" type="button" id="logoutBtn">Logout</button>

    <p id="accountMessage" class="note">You are not logged in.</p>
</form>
```

```
    </div>
</section>

<section class="container" id="booking">
    <h2>Book an Appointment</h2>
    <p class="subtitle">Select a doctor, date, and available appointment time.</p>

    <form id="bookingForm" class="form-card">
        <label>Patient Name</label>
        <input type="text" id="patientName" required>

        <label>Choose Doctor</label>
        <select id="doctorSelect" required>
            <option value="">Select a doctor</option>
        </select>

        <div class="two-columns">
            <div>
                <label>Appointment Date</label>
                <input type="date" id="appointmentDate" required>
            </div>

            <div>
                <label>Appointment Time</label>
                <select id="appointmentTime" required>
```

```
            <option value="">Select time</option>
            <option>09:00 AM</option>
            <option>10:00 AM</option>
            <option>11:00 AM</option>
            <option>12:00 PM</option>
            <option>02:00 PM</option>
            <option>03:00 PM</option>
            <option>04:00 PM</option>
        </select>
    </div>
</div>

<label>Phone Number</label>
<input type="tel" id="patientPhone" pattern="[0-9]{10}"
       maxlength="10" placeholder="10-digit phone number" required>

<label>Reason for Visit</label>
<textarea id="visitReason" rows="3"
          placeholder="Briefly describe your reason for visiting" required></textarea>

<button class="primary-btn" type="submit">Confirm Appointment</button>
<p class="note">This is a demonstration booking system, not a real medical service.</p>
</form>

<div id="bookingMessage"></div>
```

```
</section>

<section class="container" id="checkin">
  <h2>Patient Check-in Form</h2>
  <p class="subtitle">Submit your details before your scheduled visit.</p>

  <form id="checkinForm" class="form-card">
    <label>Appointment ID</label>
    <input type="text" id="checkinId" placeholder="Enter your appointment ID" required>

    <label>Current Symptoms / Visit Notes</label>
    <textarea id="checkinNotes" rows="3" required></textarea>

    <label>Have you visited this hospital before?</label>
    <select id="previousVisit" required>
      <option value="">Select answer</option>
      <option>Yes</option>
      <option>No</option>
    </select>

    <button class="primary-btn" type="submit">Submit Check-in</button>
    <p class="note">Do not enter sensitive medical details in this demo website.</p>
  </form>

  <div id="checkinMessage"></div>
```

```
</section>

<section class="container" id="history">
  <h2>My Appointment History</h2>
  <p class="subtitle">View your upcoming and previous appointments.</p>

  <div class="table-wrap">
    <table>
      <thead>
        <tr>
          <th>Appointment ID</th>
          <th>Patient</th>
          <th>Doctor</th>
          <th>Specialty</th>
          <th>Date</th>
          <th>Time</th>
          <th>Status</th>
          <th>Actions</th>
        </tr>
      </thead>
      <tbody id="appointmentTable"></tbody>
    </table>
  </div>
</section>
```

```
<section class="container" id="feedback">
    <h2>Doctor Reviews & Feedback</h2>
    <p class="subtitle">Share your experience with our doctors.</p>

    <form id="feedbackForm" class="form-card">
        <label>Patient Name</label>
        <input type="text" id="feedbackName" required>

        <label>Select Doctor</label>
        <select id="feedbackDoctor" required>
            <option value="">Select doctor</option>
        </select>

        <label>Rating</label>
        <select id="feedbackRating" required>
            <option value="">Select rating</option>
            <option value="5">★★★★★ Excellent</option>
            <option value="4">★★★★ Very Good</option>
            <option value="3">★★★ Good</option>
            <option value="2">★★ Fair</option>
            <option value="1">★ Poor</option>
        </select>

        <label>Comments</label>
        <textarea id="feedbackComments" rows="3" required></textarea>
```

```
      <button class="primary-btn" type="submit">Submit Review</button>
    </form>

    <div id="feedbackList"></div>
  </section>

  <section class="container" id="admin">
    <h2>Hospital Admin Dashboard</h2>
    <p class="subtitle">Manage doctors, appointments, and patient feedback.</p>

    <form id="adminLoginForm" class="form-card">
      <h3>Admin Login</h3>

      <label>Email</label>
      <input type="email" id="adminEmail" required>

      <label>Password</label>
      <input type="password" id="adminPassword" required>

      <button class="primary-btn" type="submit">Admin Login</button>
      <p class="note">Demo login: admin@medcare.com / med123</p>
      <p id="adminMessage"></p>
    </form>
```

```
<div id="adminDashboard" class="hidden">
  <div class="stats">
    <div class="stat-card">
      <span>Total Appointments</span>
      <h3 id="totalAppointments">0</h3>
    </div>

    <div class="stat-card">
      <span>Upcoming Appointments</span>
      <h3 id="upcomingAppointments">0</h3>
    </div>

    <div class="stat-card">
      <span>Patient Reviews</span>
      <h3 id="totalReviews">0</h3>
    </div>
  </div>

  <form id="addDoctorForm" class="form-card">
    <h3>Add Doctor</h3>

    <label>Doctor Name</label>
    <input type="text" id="newDoctorName" required>

    <label>Specialty</label>
```

```
        <select id="newSpecialty" required>
            <option>General Physician</option>
            <option>Cardiologist</option>
            <option>Dermatologist</option>
            <option>Pediatrician</option>
            <option>Dentist</option>
            <option>Orthopedic</option>
        </select>

        <label>Location</label>
        <select id="newLocation" required>
            <option>Chennai</option>
            <option>Coimbatore</option>
            <option>Bengaluru</option>
            <option>Hyderabad</option>
        </select>

        <label>Consultation Fee (₹)</label>
        <input type="number" id="newFee" min="1" required>

        <label>Available Days</label>
        <input type="text" id="newDays" placeholder="Example: Monday, Wednesday"
required>

        <button class="primary-btn" type="submit">Add Doctor</button>
    </form>
```

```
<h3>Manage Doctors</h3>
<div class="table-wrap">
  <table>
    <thead>
      <tr>
        <th>Doctor</th>
        <th>Specialty</th>
        <th>Location</th>
        <th>Fee</th>
        <th>Action</th>
      </tr>
    </thead>
    <tbody id="adminDoctorTable"></tbody>
  </table>
</div>

<h3 class="subheading">Manage Appointments</h3>
<div class="table-wrap">
  <table>
    <thead>
      <tr>
        <th>ID</th>
        <th>Patient</th>
        <th>Doctor</th>
```

```
                    <th>Date</th>
                    <th>Time</th>
                    <th>Status</th>
                    <th>Action</th>
                </tr>
            </thead>
            <tbody id="adminAppointmentTable"></tbody>
        </table>
    </div>

    <div class="admin-actions">
        <button class="secondary-btn" type="button" id="downloadReport">
            Download Appointment Report
        </button>

        <button class="secondary-btn" type="button" id="adminLogout">
            Admin Logout
        </button>
    </div>
  </div>
</section>

<footer>
  <h3>✚ MedCare Hospital</h3>
  <p>Compassionate care for a healthier tomorrow.</p>
```

```
    <p>Emergency? Contact your local emergency service.</p>

    <p>© 2026 MedCare | Internet Programming Lab Project</p>

</footer>



<script src="hospital.js"></script>

</body>

</html>
```

CSS:

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

html {
    scroll-behavior: smooth;
}

body {
    font-family: Arial, sans-serif;
    background: #f2f8fc;
    color: #304456;
    line-height: 1.6;
}

header {
    background: #075985;
    color: white;
    padding: 16px 6%;
    display: flex;
    align-items: center;
    justify-content: space-between;
    flex-wrap: wrap;
    gap: 15px;
}

.logo {
    color: #b9f2e6;
    font-size: 25px;
    font-weight: bold;
```

```
}

nav {
  display: flex;
  flex-wrap: wrap;
  gap: 15px;
}

nav a {
  color: white;
  text-decoration: none;
  font-size: 14px;
}

nav a:hover {
  color: #b9f2e6;
}

.hero {
  min-height: 360px;
  display: flex;
  align-items: center;
  padding: 55px 8%;
  color: white;
  background: linear-gradient(120deg, #075985, #0e7490, #38bdf8);
}

.hero-content {
  max-width: 650px;
}

.tag {
  color: #b9f2e6;
  font-size: 13px;
  font-weight: bold;
  letter-spacing: 2px;
}

.hero h1 {
  font-size: clamp(34px, 5vw, 50px);
  line-height: 1.15;
  margin: 12px 0;
}

.hero p {
  margin-bottom: 20px;
}
```

```
.hero-btn {
    display: inline-block;
    padding: 12px 22px;
    background: #b9f2e6;
    color: #075985;
    text-decoration: none;
    border-radius: 6px;
    font-weight: bold;
}

.hero-btn:hover {
    background: white;
}

.container {
    width: 90%;
    max-width: 1200px;
    margin: 35px auto;
    padding: 28px;
    background: white;
    border-radius: 12px;
    box-shadow: 0 4px 16px rgba(7, 89, 133, 0.08);
}

.container h2 {
    color: #075985;
    margin-bottom: 5px;
}

.subtitle {
    color: #7b8b99;
    margin-bottom: 22px;
}

.search-grid {
    display: grid;
    grid-template-columns: repeat(2, minmax(0, 1fr));
    gap: 16px;
    align-items: end;
}

label {
    display: block;
    margin: 12px 0 6px;
    font-weight: bold;
    font-size: 14px;
    color: #304456;
```

```
}

input,
select,
textarea {
    width: 100%;
    padding: 11px;
    border: 1px solid #cbdbe5;
    border-radius: 6px;
    background: white;
    font-family: Arial, sans-serif;
    font-size: 14px;
    color: #304456;
}

input:focus,
select:focus,
textarea:focus {
    outline: none;
    border-color: #0e7490;
    box-shadow: 0 0 0 3px rgba(14, 116, 144, 0.12);
}

.primary-btn,
.secondary-btn,
.danger-btn {
    padding: 11px 17px;
    margin-top: 14px;
    border: none;
    border-radius: 6px;
    cursor: pointer;
    font-weight: bold;
}

.primary-btn {
    background: #087e8b;
    color: white;
}

.primary-btn:hover {
    background: #05636d;
}

.secondary-btn {
    background: #c9f0e7;
    color: #075985;
}
```

```
.secondary-btn:hover {
   background: #a8e3d7;
}

.danger-btn {
   background: #dc3545;
   color: white;
}

.two-columns {
   display: grid;
   grid-template-columns: 1fr 1fr;
   gap: 20px;
}

.form-card {
   padding: 22px;
   margin: 16px 0;
   background: #f9fcff;
   border: 1px solid #dbeaf3;
   border-radius: 10px;
}

.form-card h3 {
   color: #075985;
   margin-bottom: 12px;
}

.note {
   margin-top: 10px;
   font-size: 12px;
   color: #7b8b99;
}

.doctor-grid {
   display: grid;
   grid-template-columns: repeat(3, minmax(0, 1fr));
   gap: 18px;
   margin-top: 22px;
}

.doctor-card {
   padding: 20px;
   border: 1px solid #d5e8f1;
   border-top: 4px solid #0e7490;
   border-radius: 10px;
   background: #f8fcff;
}
```

```
.doctor-avatar {
   width: 55px;
   height: 55px;
   border-radius: 50%;
   background: #d6f4ed;
   color: #075985;
   display: flex;
   align-items: center;
   justify-content: center;
   font-size: 23px;
   font-weight: bold;
   margin-bottom: 12px;
}

.doctor-card h3 {
   color: #075985;
   margin-bottom: 6px;
}

.doctor-card p {
   margin: 5px 0;
   font-size: 14px;
}

.fee {
   color: #087e8b;
   font-size: 20px;
   font-weight: bold;
   margin-top: 10px;
}

.table-wrap {
   width: 100%;
   overflow-x: auto;
   margin: 15px 0 25px;
}

table {
   width: 100%;
   min-width: 800px;
   border-collapse: collapse;
}

th {
   padding: 12px;
   background: #075985;
   color: white;
```

```
    text-align: left;
}

td {
    padding: 11px;
    border-bottom: 1px solid #e0ebf2;
    font-size: 13px;
}

tr:hover td {
    background: #f2faff;
}

.small-btn {
    border: none;
    border-radius: 5px;
    padding: 7px 10px;
    margin: 2px;
    cursor: pointer;
    background: #0e7490;
    color: white;
}

.small-danger {
    background: #dc3545;
}

.status-booked {
    color: #16803c;
    font-weight: bold;
}

.status-cancelled {
    color: #d62828;
    font-weight: bold;
}

.status-completed {
    color: #075985;
    font-weight: bold;
}

.message {
    padding: 15px;
    margin-top: 15px;
    background: #e7f8ef;
    border: 1px solid #b7e4c7;
    color: #17643b;
```

```
    border-radius: 8px;
}

.feedback-item {
    padding: 16px;
    margin-top: 12px;
    background: #f0f9ff;
    border-left: 4px solid #0e7490;
    border-radius: 6px;
}

.feedback-item h4 {
    color: #075985;
}

.stats {
    display: grid;
    grid-template-columns: repeat(3, minmax(0, 1fr));
    gap: 16px;
    margin: 22px 0;
}

.stat-card {
    padding: 20px;
    border-radius: 9px;
    background: #e0f2fe;
    border-left: 5px solid #0284c7;
}

.stat-card:nth-child(2) {
    background: #dcfce7;
    border-left-color: #16a34a;
}

.stat-card:nth-child(3) {
    background: #f3e8ff;
    border-left-color: #9333ea;
}

.stat-card span {
    color: #52677a;
    font-size: 13px;
}

.stat-card h3 {
    color: #075985;
    font-size: 28px;
    margin-top: 5px;
```

```
}

.subheading {
  margin-top: 25px;
}

.admin-actions {
  display: flex;
  flex-wrap: wrap;
  gap: 12px;
  margin: 15px 0;
}

.hidden {
  display: none !important;
}

footer {
  background: #075985;
  color: white;
  text-align: center;
  padding: 28px 15px;
  margin-top: 40px;
}

footer h3 {
  color: #b9f2e6;
}

footer p {
  font-size: 13px;
  margin-top: 6px;
}

@media (max-width: 850px) {
  header {
    flex-direction: column;
    align-items: flex-start;
  }

  .doctor-grid {
    grid-template-columns: repeat(2, minmax(0, 1fr));
  }

  .container {
    width: 94%;
    padding: 20px;
  }
```

```
}

@media (max-width: 550px) {
  .doctor-grid,
  .two-columns,
  .search-grid,
  .stats {
    grid-template-columns: 1fr;
  }

  nav {
    gap: 10px;
  }

  .hero {
    padding: 40px 6%;
  }

  .container {
    padding: 16px;
  }
}
```

JAVASCRIPT:

```
// MEDCARE HOSPITAL APPOINTMENT BOOKING SYSTEM
// College lab demonstration using browser localStorage.

const defaultDoctors = [
  {
    id: 1,
    name: "Dr. Arjun Kumar",
    specialty: "General Physician",
    location: "Chennai",
    fee: 500,
    days: "Monday to Saturday"
  },
  {
    id: 2,
    name: "Dr. Priya Sharma",
    specialty: "Cardiologist",
    location: "Coimbatore",
    fee: 900,
    days: "Monday, Wednesday, Friday"
  },
  {
    id: 3,
```

```
        name: "Dr. Meera Rao",
        specialty: "Dermatologist",
        location: "Bengaluru",
        fee: 700,
        days: "Tuesday to Saturday"
    },
    {
        id: 4,
        name: "Dr. Karthik Raj",
        specialty: "Pediatrician",
        location: "Hyderabad",
        fee: 600,
        days: "Monday to Friday"
    },
    {
        id: 5,
        name: "Dr. Ananya Das",
        specialty: "Dentist",
        location: "Chennai",
        fee: 400,
        days: "Monday to Saturday"
    },
    {
        id: 6,
        name: "Dr. Rahul Verma",
        specialty: "Orthopedic",
        location: "Coimbatore",
        fee: 800,
        days: "Tuesday, Thursday, Saturday"
    }
];

function loadData(key, fallback) {
    try {
        const value = localStorage.getItem(key);
        return value ? JSON.parse(value) : fallback;
    } catch (error) {
        return fallback;
    }
}

let doctors = loadData("medcareDoctors", defaultDoctors);
let patients = loadData("medcarePatients", []);
let appointments = loadData("medcareAppointments", []);
let feedbacks = loadData("medcareFeedback", []);
let checkins = loadData("medcareCheckins", []);
```

```
let currentPatient = null;
let isAdmin = false;

function saveData() {
   localStorage.setItem("medcareDoctors", JSON.stringify(doctors));
   localStorage.setItem("medcarePatients", JSON.stringify(patients));
   localStorage.setItem("medcareAppointments", JSON.stringify(appointments));
   localStorage.setItem("medcareFeedback", JSON.stringify(feedbacks));
   localStorage.setItem("medcareCheckins", JSON.stringify(checkins));
}

function escapeHTML(value) {
   return String(value).replace(/[&<>"']/g, function (char) {
      return {
         "&": "&amp;",
         "<": "&lt;",
         ">": "&gt;",
         "": "&quot;",
         "'": "&#039;"
      }[char];
   });
}

function money(value) {
   return "₹" + Number(value).toLocaleString("en-IN");
}

function todayString() {
   const now = new Date();
   return new Date(now.getTime() - now.getTimezoneOffset() * 60000)
      .toISOString().split("T")[0];
}

// Set the earliest appointment date to today.
const dateInput = document.getElementById("appointmentDate");
dateInput.min = todayString();
dateInput.value = todayString();

// SHOW DOCTOR CARDS
function doctorCard(doctor) {
   return `
      <div class="doctor-card">
         <div class="doctor-avatar">+</div>
         <h3>${escapeHTML(doctor.name)}</h3>
         <p><strong>${escapeHTML(doctor.specialty)}</strong></p>
         <p>Location: ${escapeHTML(doctor.location)}</p>
         <p>Available: ${escapeHTML(doctor.days)}</p>
```

```
        <p class="fee">${money(doctor.fee)}</p>
        <button class="primary-btn"
          onclick="chooseDoctor(${doctor.id})">
          Book Appointment
        </button>
      </div>`;
}

function renderDoctors() {
  document.getElementById("allDoctors").innerHTML =
    doctors.length
      ? doctors.map(doctorCard).join("")
      : "<p>No doctors available.</p>";

  const select = document.getElementById("doctorSelect");
  const previous = select.value;

  select.innerHTML = '<option value="">Select a doctor</option>';

  doctors.forEach(function (doctor) {
    select.innerHTML += `
      <option value="${doctor.id}">
        ${escapeHTML(doctor.name)} - ${escapeHTML(doctor.specialty)}
      </option>`;
  });

  if (doctors.some(function (doctor) {
    return String(doctor.id) === previous;
  })) {
    select.value = previous;
  }

  const feedbackSelect = document.getElementById("feedbackDoctor");
  const oldValue = feedbackSelect.value;

  feedbackSelect.innerHTML = '<option value="">Select doctor</option>';

  doctors.forEach(function (doctor) {
    feedbackSelect.innerHTML += `
      <option value="${doctor.id}">
        ${escapeHTML(doctor.name)}
      </option>`;
  });

  if (doctors.some(function (doctor) {
    return String(doctor.id) === oldValue;
  })) {
    feedbackSelect.value = oldValue;
```

```
    }
}

function chooseDoctor(id) {
    document.getElementById("doctorSelect").value = String(id);

    document.getElementById("booking").scrollIntoView({
        behavior: "smooth"
    });
}

// SEARCH DOCTORS
document.getElementById("searchForm").addEventListener("submit", function (event) {
    event.preventDefault();

    const specialty = document.getElementById("searchSpecialty").value;
    const location = document.getElementById("searchLocation").value;
    const name = document.getElementById("searchName").value.trim().toLowerCase();

    const results = doctors.filter(function (doctor) {
        return (!specialty || doctor.specialty === specialty)
            && (!location || doctor.location === location)
            && (!name || doctor.name.toLowerCase().includes(name));
    });

    document.getElementById("doctorResults").innerHTML =
        results.length
            ? results.map(doctorCard).join("")
            : "<p>No matching doctors found. Try another search.</p>";

    document.getElementById("doctorResults").scrollIntoView({
        behavior: "smooth",
        block: "nearest"
    });
});

// PATIENT REGISTRATION
document.getElementById("registerForm").addEventListener("submit", function (event) {
    event.preventDefault();

    const name = document.getElementById("regName").value.trim();
    const email = document.getElementById("regEmail").value.trim().toLowerCase();
    const password = document.getElementById("regPassword").value;

    if (patients.some(function (patient) {
        return patient.email === email;
    })) {
        alert("An account with this email already exists.");
```

```
        return;
    }

    patients.push({
        name: name,
        email: email,
        password: password
    });

    saveData();
    this.reset();

    alert("Registration successful. Please log in.");
});

// PATIENT LOGIN
document.getElementById("loginForm").addEventListener("submit", function (event) {
    event.preventDefault();

    const email = document.getElementById("loginEmail").value.trim().toLowerCase();
    const password = document.getElementById("loginPassword").value;

    const patient = patients.find(function (item) {
        return item.email === email && item.password === password;
    });

    if (!patient) {
        alert("Invalid login details. Please register first.");
        return;
    }

    currentPatient = patient;

    document.getElementById("patientName").value = patient.name;
    document.getElementById("feedbackName").value = patient.name;
    document.getElementById("accountMessage").textContent =
        "Logged in as " + patient.name;

    renderAppointments();

    alert("Login successful!");
});

// PATIENT LOGOUT
document.getElementById("logoutBtn").addEventListener("click", function () {
    currentPatient = null;

    document.getElementById("accountMessage").textContent =
        "You are not logged in.";
```

```
    document.getElementById("bookingForm").reset();
    document.getElementById("checkinForm").reset();
    document.getElementById("feedbackForm").reset();

    renderAppointments();

    alert("You have logged out.");
});

// CHECK WHETHER A DOCTOR'S TIME SLOT IS BOOKED
function slotTaken(doctorId, date, time) {
    return appointments.some(function (appointment) {
        return Number(appointment.doctorId) === Number(doctorId)
            && appointment.date === date
            && appointment.time === time
            && appointment.status === "Booked";
    });
}

// BOOK AN APPOINTMENT
document.getElementById("bookingForm").addEventListener("submit", function (event) {
    event.preventDefault();

    if (!currentPatient) {
        alert("Please register and log in before booking.");
        return;
    }

    const doctorId = Number(document.getElementById("doctorSelect").value);
    const doctor = doctors.find(function (item) {
        return item.id === doctorId;
    });

    const date = document.getElementById("appointmentDate").value;
    const time = document.getElementById("appointmentTime").value;
    const phone = document.getElementById("patientPhone").value.trim();
    const reason = document.getElementById("visitReason").value.trim();

    if (!doctor) {
        alert("Please select a valid doctor.");
        return;
    }

    if (date < todayString()) {
        alert("Please select today or a future date.");
        return;
    }
```

```
if (slotTaken(doctorId, date, time)) {
    alert("This doctor is already booked at that time. Please select another slot.");
    return;
}

const appointment = {
    id: "MC" + Date.now().toString().slice(-8),
    email: currentPatient.email,
    patient: document.getElementById("patientName").value.trim(),
    doctorId: doctor.id,
    doctorName: doctor.name,
    specialty: doctor.specialty,
    location: doctor.location,
    date: date,
    time: time,
    phone: phone,
    reason: reason,
    fee: doctor.fee,
    status: "Booked"
};

appointments.push(appointment);
saveData();

document.getElementById("bookingMessage").innerHTML = `
    <div class="message">
        <h3>Appointment Booked Successfully!</h3>
        <p><strong>Appointment ID:</strong> ${escapeHTML(appointment.id)}</p>
        <p><strong>Patient:</strong> ${escapeHTML(appointment.patient)}</p>
        <p><strong>Doctor:</strong> ${escapeHTML(appointment.doctorName)}</p>
        <p><strong>Specialty:</strong> ${escapeHTML(appointment.specialty)}</p>
        <p><strong>Date:</strong> ${escapeHTML(appointment.date)}</p>
        <p><strong>Time:</strong> ${escapeHTML(appointment.time)}</p>
        <p><strong>Consultation Fee:</strong> ${money(appointment.fee)}</p>
        <p><strong>Status:</strong> Booked (Demo)</p>
        <p class="note">This confirmation is for a college project only.</p>
    </div>`;

this.reset();
dateInput.min = todayString();
dateInput.value = todayString();

renderAppointments();
renderAdmin();

document.getElementById("bookingMessage").scrollIntoView({
    behavior: "smooth"
```

```
    });
});

// PATIENT APPOINTMENT HISTORY
function renderAppointments() {
    const table = document.getElementById("appointmentTable");

    const list = currentPatient
        ? appointments.filter(function (item) {
            return item.email === currentPatient.email;
        })
        : [];

    if (list.length === 0) {
        table.innerHTML =
            '<tr><td colspan="8">Log in to view your appointment history.</td></tr>';
        return;
    }

    table.innerHTML = list.map(function (item) {
        let statusClass = "status-booked";

        if (item.status === "Cancelled") {
            statusClass = "status-cancelled";
        } else if (item.status === "Completed") {
            statusClass = "status-completed";
        }

        let action = "—";

        if (item.status === "Booked") {
            action = `
                <button class="small-btn small-danger"
                    onclick="cancelAppointment('${item.id}')">
                    Cancel
                </button>`;
        }

        return `
            <tr>
                <td>${escapeHTML(item.id)}</td>
                <td>${escapeHTML(item.patient)}</td>
                <td>${escapeHTML(item.doctorName)}</td>
                <td>${escapeHTML(item.specialty)}</td>
                <td>${escapeHTML(item.date)}</td>
                <td>${escapeHTML(item.time)}</td>
                <td class="${statusClass}">${escapeHTML(item.status)}</td>
                <td>${action}</td>
```

```
            </tr>`;
    }).join("");
}

// CANCEL APPOINTMENT
function cancelAppointment(id) {
    if (!currentPatient) {
        alert("Please log in first.");
        return;
    }

    const appointment = appointments.find(function (item) {
        return item.id === id && item.email === currentPatient.email;
    });

    if (!appointment || appointment.status !== "Booked") {
        alert("This appointment cannot be cancelled.");
        return;
    }

    if (!confirm("Are you sure you want to cancel this appointment?")) {
        return;
    }

    appointment.status = "Cancelled";

    saveData();
    renderAppointments();
    renderAdmin();

    alert("Appointment cancelled successfully.");
}

// PATIENT CHECK-IN
document.getElementById("checkinForm").addEventListener("submit", function (event) {
    event.preventDefault();

    if (!currentPatient) {
        alert("Please log in before checking in.");
        return;
    }

    const id = document.getElementById("checkinId").value.trim();

    const appointment = appointments.find(function (item) {
        return item.id === id
            && item.email === currentPatient.email
            && item.status === "Booked";
```

```
    });

    if (!appointment) {
        alert("No active appointment was found for this ID.");
        return;
    }

    const existing = checkins.find(function (item) {
        return item.appointmentId === id;
    });

    if (existing) {
        alert("A check-in has already been submitted for this appointment.");
        return;
    }

    checkins.push({
        appointmentId: id,
        email: currentPatient.email,
        notes: document.getElementById("checkinNotes").value.trim(),
        previousVisit: document.getElementById("previousVisit").value
    });

    saveData();
    this.reset();

    document.getElementById("checkinMessage").innerHTML =
        '<div class="message">Check-in submitted successfully.</div>';
});

// PATIENT FEEDBACK
document.getElementById("feedbackForm").addEventListener("submit", function (event) {
    event.preventDefault();

    if (!currentPatient) {
        alert("Please log in before submitting feedback.");
        return;
    }

    const doctorId = Number(document.getElementById("feedbackDoctor").value);
    const doctor = doctors.find(function (item) {
        return item.id === doctorId;
    });

    if (!doctor) {
        alert("Please select a valid doctor.");
        return;
    }
```

```
    feedbacks.push({
        email: currentPatient.email,
        name: document.getElementById("feedbackName").value.trim(),
        doctorId: doctor.id,
        doctorName: doctor.name,
        rating: Number(document.getElementById("feedbackRating").value),
        comments: document.getElementById("feedbackComments").value.trim()
    });

    saveData();
    this.reset();

    renderFeedback();
    renderAdmin();

    alert("Thank you for your feedback!");
});

function renderFeedback() {
    const container = document.getElementById("feedbackList");

    if (feedbacks.length === 0) {
        container.innerHTML = "<p>No reviews have been submitted yet.</p>";
        return;
    }

    container.innerHTML = "<h3>Patient Reviews</h3>" +
        feedbacks.slice().reverse().map(function (item) {
            return `
                <div class="feedback-item">
                    <h4>${escapeHTML(item.doctorName)}</h4>
                    <p>Patient: ${escapeHTML(item.name)}</p>
                    <p>${"★".repeat(item.rating)}${"☆".repeat(5 - item.rating)}</p>
                    <p>${escapeHTML(item.comments)}</p>
                </div>`;
        }).join("");
}

// ADMIN LOGIN
document.getElementById("adminLoginForm").addEventListener("submit", function (event) {
    event.preventDefault();

    const email = document.getElementById("adminEmail").value.trim();
    const password = document.getElementById("adminPassword").value;

    if (email === "admin@medcare.com" && password === "med123") {
        isAdmin = true;
```

```
        document.getElementById("adminDashboard").classList.remove("hidden");
        document.getElementById("adminLoginForm").classList.add("hidden");

        renderAdmin();
    } else {
        document.getElementById("adminMessage").textContent =
            "Invalid admin email or password.";
    }
});

// ADMIN LOGOUT
document.getElementById("adminLogout").addEventListener("click", function () {
    isAdmin = false;

    document.getElementById("adminDashboard").classList.add("hidden");
    document.getElementById("adminLoginForm").classList.remove("hidden");
    document.getElementById("adminPassword").value = "";
});

// ADMIN ADDS A DOCTOR
document.getElementById("addDoctorForm").addEventListener("submit", function (event) {
    event.preventDefault();

    if (!isAdmin) {
        alert("Please log in as admin.");
        return;
    }

    const name = document.getElementById("newDoctorName").value.trim();

    if (doctors.some(function (doctor) {
        return doctor.name.toLowerCase() === name.toLowerCase();
    })) {
        alert("This doctor is already listed.");
        return;
    }

    const fee = Number(document.getElementById("newFee").value);

    if (fee <= 0) {
        alert("Enter a valid consultation fee.");
        return;
    }

    doctors.push({
        id: Date.now(),
        name: name,
```

```
        specialty: document.getElementById("newSpecialty").value,
        location: document.getElementById("newLocation").value,
        fee: fee,
        days: document.getElementById("newDays").value.trim()
    });

    saveData();
    this.reset();

    renderDoctors();
    renderAdmin();

    alert("Doctor added successfully.");
});

// ADMIN REMOVES A DOCTOR
function deleteDoctor(id) {
    if (!isAdmin) {
        return;
    }

    const hasActiveAppointments = appointments.some(function (item) {
        return item.doctorId === id && item.status === "Booked";
    });

    if (hasActiveAppointments) {
        alert("This doctor has active appointments and cannot be removed.");
        return;
    }

    if (!confirm("Remove this doctor from the list?")) {
        return;
    }

    doctors = doctors.filter(function (doctor) {
        return doctor.id !== id;
    });

    saveData();
    renderDoctors();
    renderAdmin();
}

// ADMIN UPDATES APPOINTMENT STATUS
function updateAppointmentStatus(id, status) {
    if (!isAdmin) {
        return;
    }
```

```
    const appointment = appointments.find(function (item) {
        return item.id === id;
    });

    if (!appointment || appointment.status !== "Booked") {
        alert("This appointment is no longer active.");
        return;
    }

    appointment.status = status;

    saveData();
    renderAppointments();
    renderAdmin();
}

// ADMIN DASHBOARD
function renderAdmin() {
    if (!isAdmin) {
        return;
    }

    const today = todayString();

    document.getElementById("totalAppointments").textContent =
        appointments.length;

    document.getElementById("upcomingAppointments").textContent =
        appointments.filter(function (item) {
            return item.status === "Booked" && item.date >= today;
        }).length;

    document.getElementById("totalReviews").textContent = feedbacks.length;

    const doctorTable = document.getElementById("adminDoctorTable");

    doctorTable.innerHTML = doctors.map(function (doctor) {
        return `
            <tr>
                <td>${escapeHTML(doctor.name)}</td>
                <td>${escapeHTML(doctor.specialty)}</td>
                <td>${escapeHTML(doctor.location)}</td>
                <td>${money(doctor.fee)}</td>
                <td>
                    <button class="small-btn small-danger"
                        onclick="deleteDoctor(${doctor.id})">
                        Remove
```

```
                </button>
            </td>
        </tr>`;
    }).join("");

    if (doctors.length === 0) {
        doctorTable.innerHTML =
            '<tr><td colspan="5">No doctors listed.</td></tr>';
    }

    const appointmentTable = document.getElementById("adminAppointmentTable");

    appointmentTable.innerHTML = appointments.map(function (item) {
        let action = "—";

        if (item.status === "Booked") {
            action = `
                <button class="small-btn"
                    onclick="updateAppointmentStatus('${item.id}', 'Completed')">
                    Complete
                </button>
                <button class="small-btn small-danger"
                    onclick="updateAppointmentStatus('${item.id}', 'Cancelled')">
                    Cancel
                </button>`;
        }

        return `
            <tr>
                <td>${escapeHTML(item.id)}</td>
                <td>${escapeHTML(item.patient)}</td>
                <td>${escapeHTML(item.doctorName)}</td>
                <td>${escapeHTML(item.date)}</td>
                <td>${escapeHTML(item.time)}</td>
                <td>${escapeHTML(item.status)}</td>
                <td>${action}</td>
            </tr>`;
    }).join("");

    if (appointments.length === 0) {
        appointmentTable.innerHTML =
            '<tr><td colspan="7">No appointments available.</td></tr>';
    }
}

// DOWNLOAD APPOINTMENT REPORT
document.getElementById("downloadReport").addEventListener("click", function () {
    if (!isAdmin) {
```

```
    alert("Please log in as admin.");
    return;
}

if (appointments.length === 0) {
    alert("No appointments available for the report.");
    return;
}

const headers = [
    "Appointment ID", "Patient", "Email", "Doctor",
    "Specialty", "Location", "Date", "Time", "Status"
];

function csvCell(value) {
    let text = String(value ?? "");

    if (/^[=+\-@\t\r]/.test(text)) {
        text = "'" + text;
    }

    return '"' + text.replace(/"/g, '""') + '"';
}

const rows = appointments.map(function (item) {
    return [
        item.id,
        item.patient,
        item.email,
        item.doctorName,
        item.specialty,
        item.location,
        item.date,
        item.time,
        item.status
    ].map(csvCell).join(",");
});

const csv = [headers.map(csvCell).join(","), ...rows].join("\r\n");
const blob = new Blob(["\uFEFF" + csv], {
    type: "text/csv;charset=utf-8;"
});

const url = URL.createObjectURL(blob);
const link = document.createElement("a");

link.href = url;
link.download = "MedCare_Appointment_Report.csv";
```

```
    link.click();

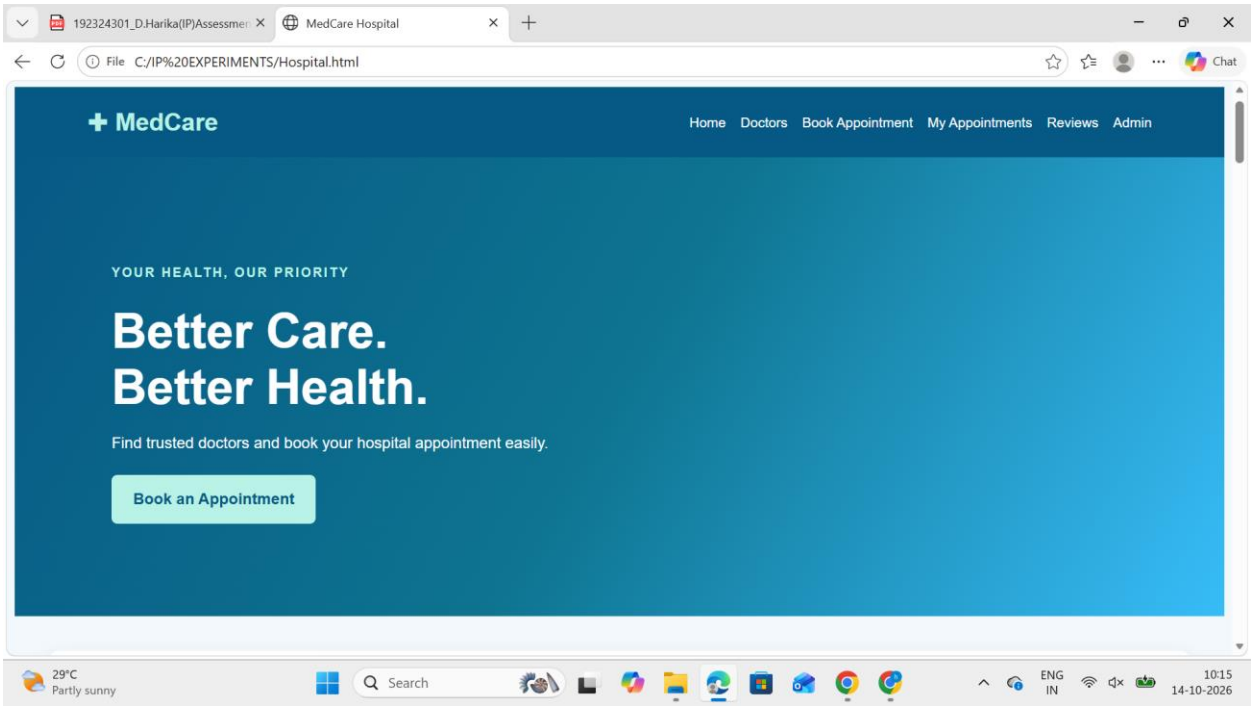
    URL.revokeObjectURL(url);
});

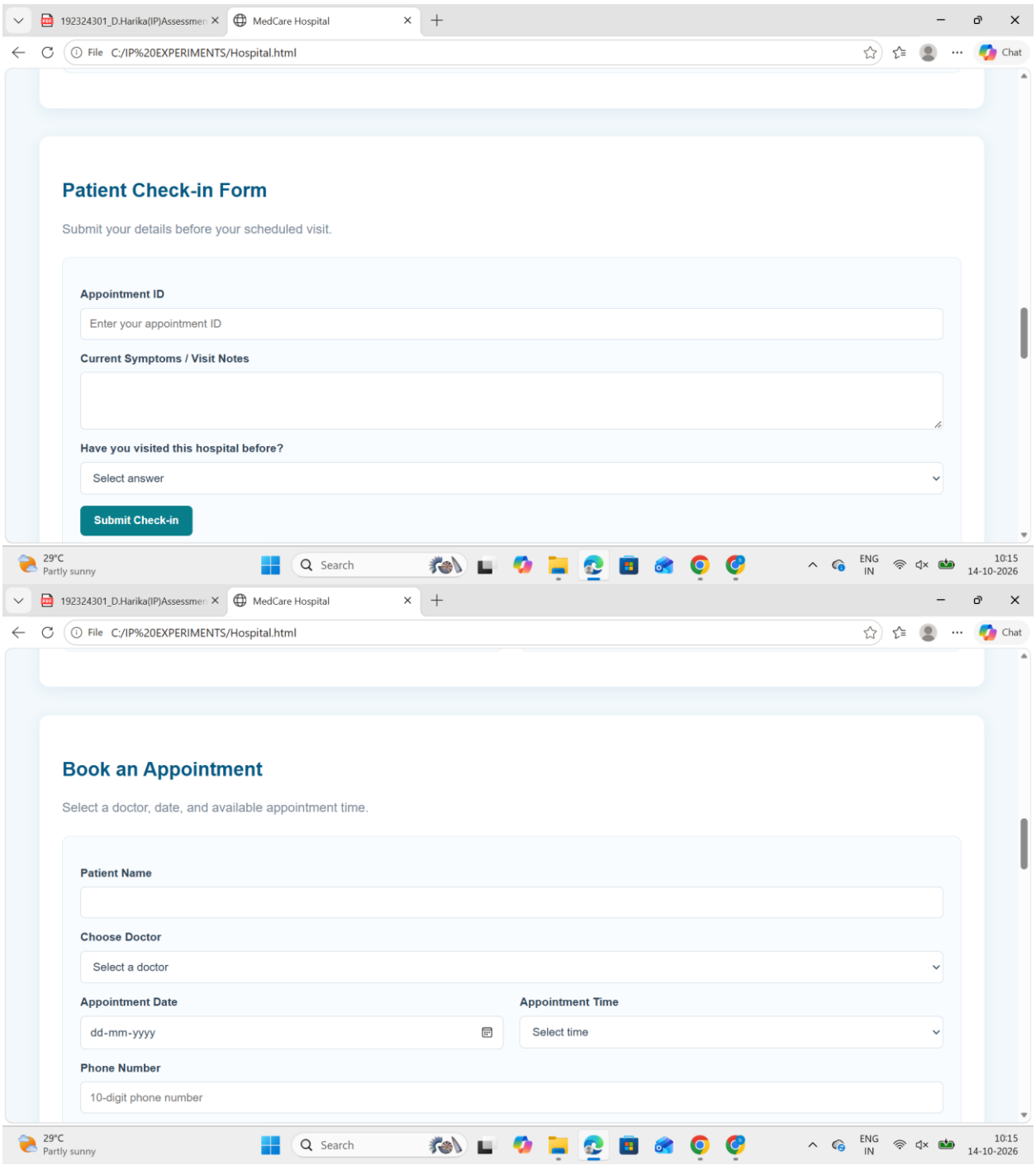
// INITIAL PAGE SETUP
renderDoctors();
renderFeedback();
renderAppointments();
```

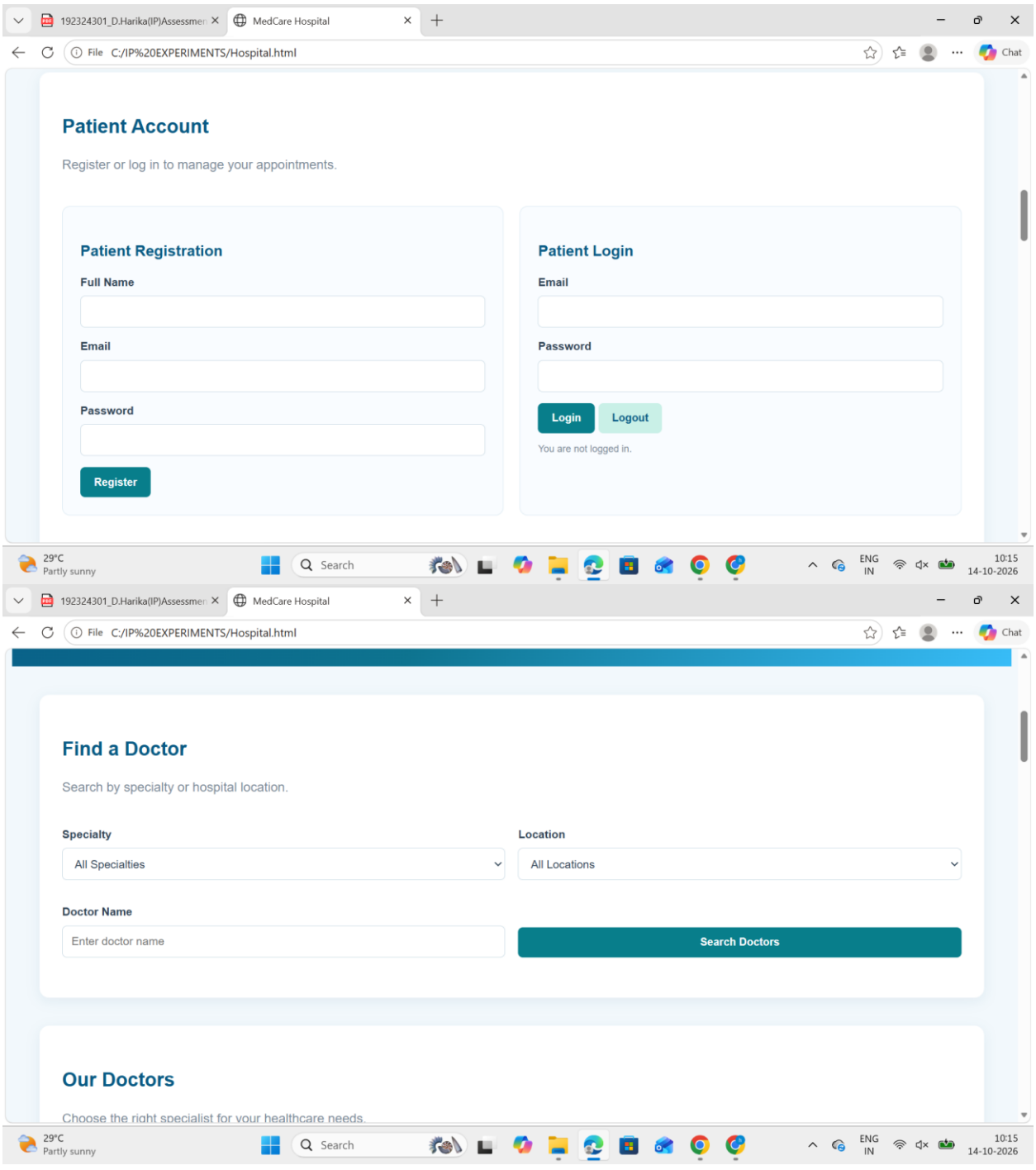
Result

The hospital appointment system was developed and tested successfully. Patients could search, book and cancel appointments and review doctors, and hospitals and the admin could manage doctors, appointments and statistics.

OUTPUT:




192324301_D.Harika(IP)Assessmen
MedCare Hospital
File C:/IP%20EXPERIMENTS/Hospital.html
Chat
Patient Check-in Form
Submit your details before your scheduled visit.
Appointment ID
Enter your appointment ID
Current Symptoms / Visit Notes
Have you visited this hospital before?
Select answer
Submit Check-in
29°C
Partly sunny
Search
ENG
IN
10:15
14-10-2026
192324301_D.Harika(IP)Assessmen
MedCare Hospital
File C:/IP%20EXPERIMENTS/Hospital.html
Chat
Book an Appointment
Select a doctor, date, and available appointment time.
Patient Name
Choose Doctor
Select a doctor
Appointment Date
dd-mm-yyyy
Appointment Time
Select time
Phone Number
10-digit phone number
29°C
Partly sunny
Search
ENG
IN
10:15
14-10-2026


192324301_D.Harika(IP)Assessmen
MedCare Hospital
File C:/IP%20EXPERIMENTS/Hospital.html
Chat
Patient Account
Register or log in to manage your appointments.
Patient Registration
Full Name
Email
Password
Register
Patient Login
Email
Password
Login
Logout
You are not logged in.
29°C
Partly sunny
Search
ENG
IN
10:15
14-10-2026
192324301_D.Harika(IP)Assessmen
MedCare Hospital
File C:/IP%20EXPERIMENTS/Hospital.html
Chat
Find a Doctor
Search by specialty or hospital location.
Specialty
All Specialties
Location
All Locations
Doctor Name
Enter doctor name
Search Doctors
Our Doctors
Choose the right specialist for your healthcare needs.
29°C
Partly sunny
Search
ENG
IN
10:15
14-10-2026